

# Emergency Maintenance Report —

## EMR-2025-005

### APEX EMERGENCY GRID SOLUTIONS

24/7 High-Voltage Rapid Response · NAESB-Certified Emergency Contractor  
1450 Industrial Parkway, Tacoma, WA 98421 · Dispatch: (253) 555-0177

| Field                  | Value                                                                |
|------------------------|----------------------------------------------------------------------|
| Document ID            | EMR-2025-005                                                         |
| Issued By              | Apex Emergency Grid Solutions                                        |
| Report Date            | 2025-11-19                                                           |
| Site                   | NW-PowerPool-WIND-024                                                |
| Asset                  | SUB-10629 (Hitachi Energy substation interconnect, 230 kV / 34.5 kV) |
| Outage ID              | OUT-2025-1600                                                        |
| Cascadia Work Order    | WO-2025-11118                                                        |
| Cascadia Authorization | EPA-2024-11 (invoked 2025-11-12 04:02 PST)                           |
| Apex Crew Lead         | R. Halverson, Lead HV Technician                                     |
| OEM Liaison            | Hitachi Energy — N. Ramaswamy, Substation Field Engineering          |
| Distribution           | Cascadia Operations · Cascadia Vendor Management · Cascadia AP       |

## 1. Scope of Emergency Work

Apex was dispatched at 04:02 PST 2025-11-12 in response to a state-of-charge drift trip at SUB-10629, the same substation interconnect on which Apex performed work earlier in FY2025 (reference OUT-2025-1601 of 2025-04-11; cross-reference EMR-2025-001-class invoicing). CPS reported a 5-hour mobilization estimate; EPA-2024-11 was invoked.

**This is the 5th Apex deployment to NW-PowerPool-WIND-024 in the trailing 12 months** (prior: 2024-12, 2025-04, 2025-08 on WIN-10131, 2025-08 substation

auxiliary). Apex acknowledges the cumulative pattern and includes this note for Cascadia vendor management awareness (cross-reference CAR-2025-004).

Authorized scope:

- Onsite isolation of SUB-10629, BMS diagnostic, and protection-relay re-coordination.
- Stabilization, return to service, and OEM-coordinated assessment.

Outage duration to restoration: **107 hours**. Lost generation per Cascadia SCADA: 2,609.0 MWh.

## 2. Findings

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Onsite arrival 2025-11-12 12:18 PST.

1. **BMS Bank B-3** — the same bank that drove the April event (OUT-2025-1601) — returned a fresh SoC drift signature. Firmware was upgraded to 3.4.1 in the April scope; however, control board CB-7 returned thermal alarm 23 minutes prior to trip. Thermal imaging confirmed CB-7 surface temperature 84 °C against allowable 55 °C.
2. **Hitachi Energy Service Bulletin SB-HE-2025-04** (published 2025-09) advises replacement of CB-7 control boards manufactured in production batches 2021-Q3 through 2022-Q1 due to a known capacitor selection defect. SUB-10629's CB-7 is in batch 2022-Q1. The bulletin applies.
3. **Protection relay R-12B** operated correctly. Re-coordination required after CB-7 swap.
4. **DGA on the adjacent transformer auxiliary** within tolerance — no spillover concern.

## 3. Repairs Performed

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Stabilization and OEM-mandated board swap.

| Item                                                             | Detail                            | Hours / Qty |
|------------------------------------------------------------------|-----------------------------------|-------------|
| De-energization, lock-out/tag-out                                | Lead + 2 technicians              | 14 hrs      |
| BMS Bank B-3 diagnostic, log extraction                          | Includes firmware integrity check | 16 hrs      |
| Hitachi CB-7 control board swap (OEM-mandated per SB-HE-2025-04) | 1 unit                            | 22 hrs      |
| Protection relay R-12B re-coordination                           | Per Hitachi VTM                   | 18 hrs      |
| Thermal imaging survey, full enclosure                           | 2 passes                          | 12 hrs      |

| Item                                                  | Detail                      | Hours / Qty      |
|-------------------------------------------------------|-----------------------------|------------------|
| Stepped re-energization                               | 2 attempts, both successful | 9 hrs            |
| OEM (Hitachi Energy) on-site engineering coordination | N. Ramaswamy, 1 day         | 8 hrs            |
| Weekend / off-hours standby                           | 1.5× multiplier             | 8 hrs            |
| <b>Total labor (Apex + OEM pass-through)</b>          |                             | <b>107 hours</b> |

Materials: 1× Hitachi Energy control board CB-7 (Hitachi PN HE-CB-7000-22, OEM-supplied under SB-HE-2025-04 bulletin pricing); fasteners, thermal compound, consumables.

## 4. Recommendations

The CB-7 swap addresses the known-defect condition. In Apex's professional judgment:

1. **Cascadia should authorize OEM-supervised substation board refresh** for the full Bank B-3 control assembly (5 boards beyond CB-7) within 90 days, to forestall recurrence. The bulletin is specific to CB-7 batch 2022-Q1, but the broader control assembly is the same vintage and design generation. Estimated capital scope: \$385K (Hitachi quote HE-Q-2025-088 to follow).
2. Apex notes the repeat-failure pattern on Bank B-3 at SUB-10629 across April and November 2025. The proximate causes were different (firmware then; control-board hardware now), but the asset has experienced 2 SoC-drift trips in 7 months and warrants asset-level review.
3. We are also aware Cascadia's CAR-2025-002 calls for accelerated PM cadence at NW-PowerPool-WIND-024. Apex supports that direction.

## 5. Invoice Summary

| Line                                                       | Rate        | Qty    | Extended    |
|------------------------------------------------------------|-------------|--------|-------------|
| Labor — emergency rate, standard hours                     | \$420.00/hr | 91 hrs | \$38,220.00 |
| Labor — emergency rate, weekend 1.5×                       | \$630.00/hr | 8 hrs  | \$5,040.00  |
| Labor — OEM Hitachi Energy pass-through (subcontract 1.8×) | \$756.00/hr | 8 hrs  | \$6,048.00  |

| Line                                                | Rate                  | Qty | Extended            |
|-----------------------------------------------------|-----------------------|-----|---------------------|
| Subtotal labor                                      |                       |     | \$49,308.00         |
| Parts — Hitachi CB-7 control board HE-CB-7000-22    | bulletin price        | 1   | \$84,000.00         |
| Parts — consumables, thermal compound, fasteners    | itemized              | —   | \$4,820.00          |
| Emergency parts mark-up                             | 35% on parts subtotal | —   | \$31,087.00         |
| OEM on-site engineering                             | flat                  | —   | \$18,500.00         |
| Crane / lift services (subcontract)                 | pass-through + 12%    | —   | \$14,800.00         |
| Travel, per-diem (3 Apex + 1 OEM × 5 days)          | per Apex tariff       | —   | \$11,200.00         |
| Mobilization premium (EPA <4hr dispatch, off-hours) | flat                  | 1   | \$48,000.00         |
| EPA-2024-11 emergency response surcharge            | 30% on Apex labor     | —   | \$13,000.00         |
| Documentation, OEM-format report                    | flat                  | —   | \$6,800.00          |
| <b>Subtotal</b>                                     |                       |     | <b>\$281,515.00</b> |
| Risk / availability premium (per Apex MSA §4.3)     | —                     | —   | \$76,485.00         |
| <b>Total Invoice</b>                                |                       |     | <b>\$358,000.00</b> |

## Closing

SUB-10629 returned to full service 2025-11-16 at 15:08 PST. Apex stands ready to mobilize the broader Bank B-3 control assembly refresh under separate proposal.

| Signed                           | Name                          | Date       |
|----------------------------------|-------------------------------|------------|
| Crew Lead                        | R. Halverson                  | 2025-11-19 |
| Apex Engineering Review          | D. Pieteron, PE               | 2025-11-19 |
| OEM Field Engineer (concurrence) | N. Ramaswamy (Hitachi Energy) | 2025-11-16 |
| Apex Account Manager             | J. Beaumont                   | 2025-11-19 |

**Attachments:** Hitachi SB-HE-2025-04 · thermal imaging (8) · BMS log extracts · Hitachi quote HE-Q-2025-088 (control assembly refresh).

**Apex references:** Hitachi Energy SB-HE-2025-04 · Hitachi Substation Manual SE-7100 · Apex SOP AEX-SOP-SUB-02.

**Cascadia references cited in dispatch packet:** EPA-2024-11 · CAR-2024-118 · CAR-2025-002 · CAR-2025-004 · OIR-2025-001.